

James Smith

Embedded Systems & IoT Firmware Engineer • 5 Years Experience • MERIQ Index: 72% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: MQTT, C/C++, I2C/SPI/UART, Python, ARM Cortex, PCB Debugging, Embedded Linux, FreeRTOS

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core MQTT service, reducing latency by 27%.
- Led a squad of 5 engineers delivering scalable microservices processing 33M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 59%.

Associate Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2021 - 2023)

- Designed and maintained C/C++ data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using I2C/SPI/UART saving \$24k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [MQTT, C/C++, I2C/SPI/UART, Python]

Engineered end-to-end production solution leveraging MQTT and C/C++ with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [I2C/SPI/UART, Python, ARM Cortex, PCB Debugging]

Architected high-resilience system integrating I2C/SPI/UART with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Cybersecurity and Information Assurance

University of Waterloo — 2017 - 2021 (Score: 3.69 / 4.0)